

- 4 Respiratory syncytial virus (RSV) can infect the human gas exchange system and cause disease in human populations.

(a) Explain why the disease caused by RSV is an example of an infectious disease.

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..... [2]

(b) Figure 4.1 shows the structure of RSV.

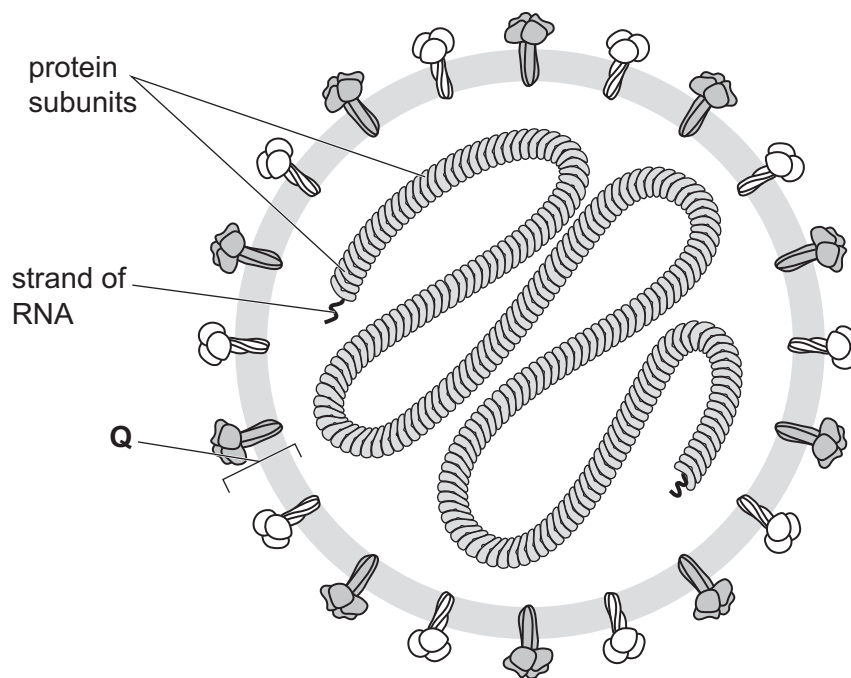


Figure 4.1

State the name of:

- the outer layer labelled **Q** in Figure 4.1

.....

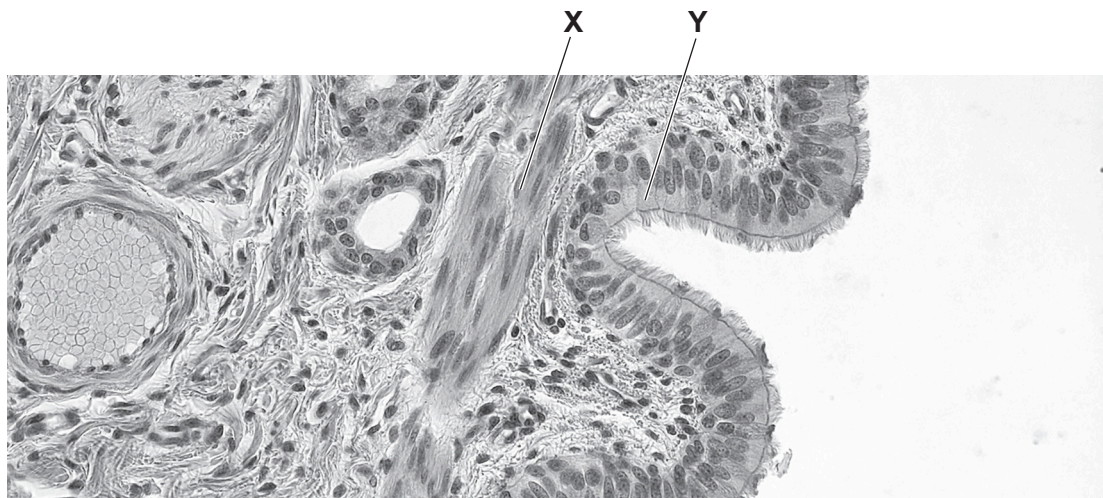
- the structure in Figure 4.1 composed of protein subunits.

.....

[2]

- (c) RSV can infect tissues of the bronchus.

Figure 4.2 is a photomicrograph of tissues in a bronchus.



magnification $\times 500$

Figure 4.2

- (i) State the name of the tissue labelled **X** in Figure 4.2.

..... [1]

- (ii) State the function of the cell labelled **Y** in Figure 4.2.

..... [1]

- (d) There are two methods that can be used to prevent people from becoming ill if they come into contact with RSV. These methods are:

- injection with a monoclonal antibody
- vaccination.

- (i) State the forms of immunity that people would be expected to develop as a result of each of these two methods.

Use **two** words to describe each form of immunity.

injection with a monoclonal antibody

..... and immunity

vaccination

..... and immunity

[2]



- (ii) Outline how a monoclonal antibody to protect people against RSV is produced by the hybridoma method.

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..... [4]

- (iii) State **one** advantage of using a vaccine, instead of a monoclonal antibody, to protect against RSV infection.

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..... [1]

- (e) Phagocytes can remove cells in the body that have been infected with viruses such as RSV.

- (i) Outline the events that lead to the formation of a phagocytic vacuole after detection of the infected body cells by a phagocyte.

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..... [2]

- (ii) State the name of the organelle in a phagocyte that contains enzymes to destroy the infected cells.

..... [1]

[Total: 16]